

HANG ZHANG

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SUMMARY

Research and engineering leader in multimodal foundation models and physical AI. Lead XPeng's VLA Foundation Model team of approximately 45 researchers and engineers, taking models from architecture design to production. Previously Senior Staff at Cruise and Research Scientist at Meta. Research impact: 12,000+ citations and 10,000+ GitHub stars.

Research interests: Multimodal foundation models; vision-language-action models; physical AI; world models; long-context architectures.

EXPERIENCE

XPeng

Senior Principal Engineer / Senior Director

Dec 2024 – Present

Silicon Valley, CA

Head of VLA Foundation Model

- Started as an **individual contributor**, building **XPeng VLA 2.0** from scratch. Progressively built and scaled the team to approximately **45 researchers and engineers** across Silicon Valley and China, taking VLA from **prototype to production in approximately one year**.
- Now lead model architecture, training, optimization, and production integration. The pure-vision, end-to-end VLA model powers driver assistance across **300+ Chinese cities**, overseas vehicles, and Robotaxi.
- Delivered three consecutive 2026 releases that roughly **halved key safety and intervention metrics** across highway, urban, and parking scenarios, while increasing VLA mileage penetration by **over 50%**.
- Designed **Infini VLA**, a streaming VLA architecture shipped in 2026. Combined LLM-style long-sequence training with **autoregressive streaming inference over a temporal KV cache**, separating long-horizon visual memory from immediate reaction on unchanged on-vehicle compute. **Wrote the initial implementation** and led training, optimization, and integration.
- Advised **TuringLLM**, a 20B-A2B mixture-of-experts VLM for physical AI, and **TuringViT**, advancing the team's language and vision foundation models.
- Advised **X-Mind**, a predictive world model using visual prediction as visual chain-of-thought for end-to-end driving.

Cruise AI

Senior Staff Applied Research Scientist

Sep 2022 – Dec 2024

San Francisco, CA

- Led consolidation of camera detection and segmentation models, bringing perception workloads into unified multi-task architectures.
- Led development of a **multi-view, multi-frame, multimodal perception model** spanning multiple vehicle platforms.
- Led the shared perception training pipeline supporting detection, segmentation, long-tail understanding, lane detection, off-board foundation models, and **online distillation**.

EXPERIENCE (CONTINUED)

Meta Reality Labs (Facebook)

Research Scientist

Oct 2020 – Aug 2022

Menlo Park, CA

- Led **FBNAS**, a unified platform for cross-platform, hardware-aware neural architecture optimization, applied to production models for **Instagram person understanding, AR/VR, and ads**. Supported models across Instagram, Portal, and VR devices.
- Developed and open-sourced **D2Go**, bringing Detectron2 to mobile deployment.
- Conducted research on **open-vocabulary segmentation and efficient architectures**: OV-Seg, multi-task neural architecture search (FBNetV5), and hardware-aware vision transformer scaling (ScaleViT).

Amazon AWS AI

Senior Applied Scientist

Jan 2018 – Oct 2020

East Palo Alto, CA

- Led development of **GluonCV** and **AutoGluon**, supporting computer vision and automated machine learning in the MXNet ecosystem.
- Led research on large-scale vision models and training methods, including **ResNeSt**, Bag of Tricks, CFNet, and dynamic SGD.

Earlier research: Amazon Lab126, Applied Scientist Intern (2017): EncNet semantic segmentation, CVPR 2018 oral. NVIDIA, Deep Learning Research Intern (2016): end-to-end autonomous driving and a Torch-to-Caffe model converter.

SELECTED RESEARCH

Foundation models and physical AI

TuringLLM — Efficiently Scaling Foundation Models Toward Physical AI. arXiv, 2026.

X-Mind — Visual chain-of-thought via predictive world modeling for driving. arXiv, 2026.

TuringViT — Making SOTA Vision Transformers Accessible to All. arXiv, 2026.

Vision, representations, and efficient architectures

CLAP — Unsupervised 3D representation learning for fusion 3D perception. ICLR, 2026.

OV-Seg — Open-Vocabulary Semantic Segmentation with Mask-Adapted CLIP. CVPR, 2023.

FBNetV5 — Neural Architecture Search for Multiple Tasks in One Run. arXiv, 2021.

ResNeSt — Split-Attention Networks. CVPR, 2022.

EncNet — Context Encoding for Semantic Segmentation. CVPR, 2018 (oral).

Full publication list: [Google Scholar](#) — 12,000+ citations.

EDUCATION

Rutgers University — Ph.D., Electrical and Computer Engineering

2013–2017

Advisor: Prof. Kristin Dana.

Southeast University — B.S., School of Automation

2009–2013

RECOGNITION AND OPEN SOURCE

[Top 50 Intelligent Driving Developers in China \(2025\)](#).

[10,000+ GitHub stars](#) across open-source projects; contributions include D2Go, GluonCV, and AutoGluon.